

Group-5

Group members:

1.Misha Maliha-23524202029

2. Jannatul Shanjimom Aurpy-2352420123

3. Faiza Mustari-23524202043

4. Mehar Nigar Annona -23524202147

Project Title: Sustainable Farm Decision Support and Crop Advisory System

1. Introduction

1.1 Purpose

The purpose of this system is to provide farmers with a digital platform that supports decision-making for crop selection, irrigation scheduling, fertilizer management, and crop health monitoring. The system aims to improve farm productivity, optimize resource usage, and reduce crop loss through timely recommendations.

1.2 Intended Audience

- **Primary Users:** Farmers with basic technical knowledge
- **Secondary Users:** Agriculture officers, students, and researchers interested in farm management and crop monitoring

1.3 Intended Use

The system will be used by farmers to:

- Input farm details (land size, soil type, previous crops)
- Receive crop recommendations based on season and soil
- Schedule irrigation efficiently
- Optimize fertilizer usage
- Maintain a log of crop health and disease records

1.4 Product Scope

The system is a **modular software application** that provides:

- Crop planning advisory
- Irrigation scheduling
- Fertilizer recommendation
- Disease and crop health logging
- Team-based development workflow with Git, Trello, and Discord support

The system will initially function as a **decision support platform** and can be extended in the future to include **AI-based disease detection, yield prediction, and predictive analytics** for sustainable agriculture.

1.5 Risk Definitions

Risk	Description	Mitigation
Data input errors	Incorrect farm details may lead to wrong recommendations	Validate user input forms
System downtime	Server or software failure	Regular backups and version control using Git
Team collaboration issues	Miscommunication among developers	Use Trello for task management and Discord for real-time communication
Scalability limitations	Large datasets in future AI modules	Design modular and scalable architecture

2. Overall Description

2.1 User Classes and Characteristics

User Class	Characteristics	Needs
Farmer	Basic computer skills	Easy-to-use interface, crop planning, irrigation advice
Agriculture Officer	Intermediate technical skills	Analyze farm performance, monitor multiple farms
Researcher/Student	Advanced technical knowledge	Data for research, future AI integration

2.2 User Needs

- User-friendly interface for non-technical users
- Reliable crop recommendation based on soil and season
- Accurate irrigation and fertilizer planning
- Maintainable records for crop health and disease
- Collaborative software development workflow

2.3 Operating Environment

- Platform: Desktop application (Windows/Linux)
- Database: SQLite/MySQL for storing farm and crop data
- Version Control: GitHub
- Project Management: Trello
- Team Communication: Discord

2.4 Constraints

- Users must have basic computer skills
- Initial system will rely on rule-based decision logic (AI extension is optional)
- Offline functionality may be limited; internet required for collaboration features
- Modular design must be followed for future expansion

2.5 Assumptions

- Users provide accurate and timely information about farm and crops
- System will initially support commonly cultivated crops in the target region
- Developers will follow Git branching and Trello task management for collaboration

3. Requirements

3.1 Functional Requirements

- 1. Farmer Profile Module:**
 - a. Input land size, soil type, and previous crop history
 - b. Update and maintain profile data
- 2. Crop Planner:**
 - a. Suggest crops suitable for the user's soil and season
 - b. Provide reasoning and expected productivity
- 3. Irrigation Scheduler:**
 - a. Recommend watering frequency and quantity per crop
 - b. Maintain irrigation history
- 4. Fertilizer Manager:**
 - a. Recommend fertilizer type and quantity per crop
 - b. Track fertilizer application history
- 5. Disease Log:**
 - a. Record leaf spots, yellowing, pest infestations
 - b. Generate crop health reports
- 6. Collaboration Tools Integration:**
 - a. Git for version control
 - b. Trello for task tracking
 - c. Discord for team communication

3.2 Non-Functional Requirements

Type	Requirement
Usability	Simple and intuitive user interface suitable for farmers
Reliability	System should provide accurate recommendations based on input
Performance	Minimal response time for recommendations
Maintainability	Modular code structure to allow easy updates and AI extension
Security	Protect user data with login credentials
Scalability	Support additional crops, farms, or AI modules in the future
Collaboration	Git branching, Trello task boards, and Discord communication integrated